



SEE THE FUTURE STARTUP COMPETITION 2026 · COLUMBIA CLIMATE SCHOOL

BetterBarter

A dedicated higher-education campus marketplace that connects buyers and sellers to promote item reuse, avoid unnecessary new purchases, and reduce waste — **a one-stop shop built on convenience, circularity, self-sustenance and safety.**

THE PROBLEM

Every year, predictably, the same thing happens to the same people.

Most incoming and outgoing campus residents in New York City want to buy, sell, give or receive used items — and go through fragmented channels to do it: WhatsApp group chats, Facebook Marketplace, eBay, physical thrift stores, Amazon Prime.

Poor Timing & Fit

Buyers may find a cheap item online, only to see pickup is out of reach. Sellers reach a general audience, with no way to find other campus members nearby.

No Trust or Verification

For most online transactions, community members agree to buy or sell to an unverified stranger, sometimes having to reveal their own personal details.

Waste, Waste, Waste

Buyers may end up paying more for what other community members are selling at a cheaper rate. Sellers donate or discard what they cannot sell on time.

12.1M t

US furniture and furnishings generated in 2018 — 80.1% landfilled (EPA)

17M t

US textiles the same year; 11.3M t straight to landfill (EPA)

1 week

of May carries most of a residential campus's annual discard

Meet the BetterBarter Marketplace.

A one-stop shop based on the principles of **convenience, circularity, self-sustenance and safety.**

Campus-specific Community

Buy, sell, give, or receive items exclusively from verified campus community members. Filter by distance and choose to transact with people who live near you.

Environmentally Conscious

Two campus community members crossing a campus by foot or bike. No gas vehicle, no excessive packaging. Most importantly, items get a longer “life” and avoid being discarded too soon.

Built-in Safety

Chat with other members using only your first name. No phone numbers, no emails, no personal information exchanged. Meet up on or near campus, whatever is safe and convenient for you.

Variety & Relevance

Access new or used items from textbooks and furniture to fashion and baby toys, all from people nearby. We are more alike than we think.

Deployed on a live backend, not a mockup.

List it in two steps

Up to three photos, then a short form: category, condition, size, how you are listing it, and where you want to meet — from a list of real campus places, never a room.

Only your campus sees it

One verified university email per account, enforced in the database rather than the interface. Every school is an isolated Marketplace.

Claim, then meet

Claiming opens a thread and holds the item for three hours, so two people are not walking to the same lobby. Contact details are refused at the point of sending.

The community is the logistics

A fridge is not a one-person object. A listing can ask for two people or a trolley, and a student on the same campus takes that work for a fee paid directly between them — no van dispatched, so the transport term stays zero.

Both sides confirm, in person

Six digits, split three-and-three across the two phones. Neither app can read the other's half, so the count cannot be raised by one person alone.

It keeps itself tidy

An hourly job asks whether a listing is still there and retires it if nobody answers, so a Marketplace full of things that already went cannot happen.

Realistic and simple: five numbers.

Circularity & self-sustenance

Items diverted from waste — used items given away or sold instead of thrown out or donated externally.

Number of items given for free.

Estimated CO₂e avoided per item, on the published model overleaf.

Safety

Percentage of active users with .edu verification — 100% by construction: the database will not create an account without one.

Reports per 1,000 transactions, and how many of them were actioned.

And one line on why the first number can be trusted. A handoff is not counted when somebody taps a button — it is counted when six digits, split three-and-three across the two phones, are put back together in person. Neither app can read the other's half. A reuse figure one person can raise alone is not a figure, and every other campus reuse number we can find is self-reported or weighed at the back of a truck.

Three levels, kept apart, so nobody takes our word for the third.

MEASURED

1 object

A confirmed handoff. Counted, not estimated — the only number here that is observed.

ESTIMATED

3.45 kg

Mass kept in use, from a published factor table weighted to a mix of small things rather than one carried by bikes and laptops.

ESTIMATED + ADJUSTED

7.2 kg

CO₂e avoided, at a **displacement rate of 0.5** — we assume only half of reuse events prevent a purchase.

The assumption most likely to be challenged, given away before anyone asks: a rented object earns no carbon credit at all in this model, because it comes back.

Sources: EPA *Advancing Sustainable Materials Management* (durable goods, textiles); EPA GHG Emission Factors Hub; EPA Waste Management Hierarchy; NCES IPEDS enrolment; WRAP, *Environmental and Economic Benefits of Re-use*. The factor table, the mix and the arithmetic are one file in an open repository.

The material in play, and the revenue against it.

THE MATERIAL

TAM	≈46M objects	≈333,000 t CO ₂ e
SAM	≈14M objects	≈101,000 t
SOM · yr 3	≈300,000	≈2,160 t

15.4M US undergraduates (NCES) × 3 reusable objects a year, at our published 3.45 kg and 7.2 kg CO₂e per object.

THE REVENUE

TAM	\$58M a year
SAM	\$18M a year
SOM · yr 3	\$375k a year

3,896 US degree-granting institutions (NCES 2022–23) at a \$15k licence; SAM is the ~1,200 residential four-year campuses; SOM is 25 of them. Placements and boosts excluded.

Never on a physical item. Free or not.

Transaction Fee

A 1–2% transaction fee will be applied to higher-value service categories in later phases — tutoring, babysitting, car washing. **No transaction fees will be applied to any physical item, free or not.**

Institutional Partnerships & Licensing Fees

The main line. Priced against a cost the university already carries — move-out hauling, plus the circularity reporting it has to produce anyway — and it buys object-level, auditable reuse data no collection programme can produce.

Boosted Listings

A small, affordable flat fee for sellers who wish to boost their listing, especially if it is an urgent sale.

Advertisement Fees

Charged to local businesses — food and drink, clinical services, and other industries that do not cannibalise our own — who wish to reach a hyper-targeted, verified higher-education community.

One group, then one school, then one university.

PHASE 0 · NOW

50–100 friends

Columbia SPS, by hand.
Enough people to find out what breaks before anybody official is watching.

PHASE I · 0–1 MONTH

School of Professional Studies

One school, opened properly, through the channels students already read.

PHASE II · 1–6 MONTHS

Columbia University

University-wide — and through the 2027 move-out, which is the week the whole thesis is tested.

PHASE III · 6 MONTHS ON

Other institutions

A university email domain already maps to a campus server-side, so a new school's Marketplace exists the moment its first student signs in.

Columbia's first-years moved in on 29–30 August and orientation runs to 7 September — Phase 0 is happening in the week this is being submitted, not after it.

What exists, and what does not.

What exists

A live product at betterbarter.vercel.app, open source. Row-level security on every table, an RLS-hardening pass after the security advisor flagged it, sign-up gated to academic domains server-side, rate limits in the database, a reporting and moderation queue, and a confirmation function that cannot double-count.

The impact model published in full — factor table, category mix, displacement rate, every source. All of it built and shipped by two people in five weeks.

What does not

No paying customers. No institutional partnership. No pilot users — two verified Columbia accounts exist and both of them are ours. No conversation yet with Columbia Housing, EcoReps or the Office of Sustainability.

We would rather state that plainly than dress it up. What we are asking to be judged on is a working product and a defensible method, not a user count we do not have.

A pilot campus, and people willing to argue with the model.

Agung Nugroho

CO-FOUNDER · TECH AND PRODUCT

Builds and runs the Marketplace, the impact model and the data. Graduate student and Graduate Assistant at Columbia SPS; background in customer-lifecycle and CRM marketing at telecommunications scale.

Tessa Wong

CO-FOUNDER · STRATEGY AND COMMUNITY

A sustainable development professional by background. Began her career as a management consultant at Accenture Singapore, before pivoting to sustainable development research and consulting — first for the development sector in Cambodia, then for the private sector in Singapore.

The Ask

An introduction to a Columbia housing or sustainability office, a pilot campus for the 2027 move-out, and a hard read of the displacement assumption in the impact model.